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Study Material • 2026 Edition

CHM 101

Chemical Kinetics

The complete study guide: broken-down concepts, mnemonics, worked examples and high-yield topics.

Mnemonics **Worked Examples** **Concepts Simplified** **High-Yield**

- Rate of reaction & rate expressions
- Rate law, order & rate constant
- Method of initial rates
- Integrated rate laws & half-life
- Collision theory & factors affecting rate
- Activation energy & Arrhenius equation
- Catalysis & enzymes
- Reaction mechanisms & rate-determining step

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How to use this guide

Chemical kinetics is the **“speed”** chapter of CHM 101: how fast reactions go, what controls the speed, and what is happening at the molecular level. It is calculation-heavy and loves to repeat the same question types, so the fastest way to score is to master the **10 formulas** and the **5 calculation patterns** in this guide.

Tag	What it means
★ HIGH-YIELD	Appears again and again in tests and exams. Learn these first.
MNEMONIC	A memory trick for formulas, rules or lists.
WORKED EXAMPLE	A full solution, step by step. Cover the answer and try it first.
BROKEN DOWN	A tricky idea split into small, simple steps.
EXAM TRAP	A common mistake that costs marks.

★ HIGH-YIELD **The Top 12 high-yield topics (study these in this order)**

- **Rate expressions** with stoichiometric coefficients (divide by the coefficient, watch the sign).
- **Rate law** and order: determined *by experiment only*, never from the balanced equation.
- **Units of k** for zero, first, second and third order.
- **Method of initial rates**: finding orders and k from a data table.
- **First-order kinetics**: $\ln[A] = \ln[A]_0 - kt$, and the exponential form.
- **Half-life**: formulas, how each depends on $[A]_0$, and fraction remaining = $(½)^n$.
- **Linear plots and slopes** for each order (which graph is a straight line?).
- **Collision theory**: the three conditions for an effective collision.
- **Arrhenius equation**: finding E_a from a graph or from two temperatures.
- **Energy profile diagrams** and exactly what a catalyst does (and does not do).
- **Factors affecting rate** and which of them change k.
- **Mechanisms**: molecularity vs order, rate-determining step, intermediates vs catalysts.

Contents

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1 Rate of Reaction & Rate Expressions

What "rate" means and how to write it correctly

Chemical kinetics is the study of the **speed (rate)** of chemical reactions, the factors that affect that speed, and the **mechanism** (the step-by-step path) by which reactants become products. Thermodynamics tells you *whether* a reaction can happen; kinetics tells you *how fast*. A reaction can be thermodynamically favourable yet extremely slow (diamond turning into graphite).

Rate of reaction

Rate of reaction = the change in concentration of a reactant or product per unit time. SI-style unit: **mol L^{-1} s^{-1}** (same as mol dm^{-3} s^{-1} or M s^{-1}). For slow reactions you may see mol L^{-1} min^{-1} or h^{-1}.

$$\text{Rate} = \Delta[\text{concentration}] / \Delta t$$

Reactants are **used up** so their concentration falls; products **build up** so theirs rises. Because a rate is always reported as a **positive** number, we put a **minus sign** in front of a reactant term.

BROKEN DOWN Writing the rate expression in 4 steps

- **Step 1:** Write the balanced equation: aA + bB → cC + dD.
- **Step 2:** For each species write the change in concentration over time, d[X]/dt.
- **Step 3:** Put a **−** for reactants and a **+** for products.
- **Step 4: Divide each term by its coefficient** so they all equal the same single "reaction rate".

$$\text{Rate} = -\frac{1}{a}\frac{d[A]}{dt} = -\frac{1}{b}\frac{d[B]}{dt} = +\frac{1}{c}\frac{d[C]}{dt} = +\frac{1}{d}\frac{d[D]}{dt}$$

MNEMONIC Signs and coefficients

Reactants Reduce (−), Products Pile up (+). Divide by the coefficient.

The negative sign is only there to make the rate positive. Always check the answer is positive.

Average, instantaneous and initial rate

Type	Meaning	How to get it
Average rate	Change over a *time interval*	Δ[A]/Δt between two points on the curve
Instantaneous rate	Rate at one exact moment	Slope of the **tangent** to the [A] vs t curve at that time
Initial rate	Instantaneous rate at t = 0	Tangent at t = 0 (used in the method of initial rates, Part 3)

WORKED EXAMPLE 1: Average rate

The concentration of A falls from **0.80 M** at 0 s to **0.50 M** at 30 s. Find the average rate of disappearance of A.

Solution: rate = $-\Delta[A]/\Delta t = -(0.50 - 0.80)/(30 - 0) = 0.30/30 =$ **$0.010\ M\ s^{-1}$**

WORKED EXAMPLE 2: Rates from stoichiometry

For $N_2 + 3H_2 \rightarrow 2NH_3$, NH_3 forms at **$0.60\ mol\ L^{-1}\ s^{-1}$**. Find the rate of disappearance of H_2 and N_2, and the reaction rate.

Reaction rate = $(1/2)\ d[NH_3]/dt = 0.60/2 =$ **$0.30\ mol\ L^{-1}\ s^{-1}$**

$-d[H_2]/dt = 3 \times 0.30 =$ **$0.90\ mol\ L^{-1}\ s^{-1}$** | $-d[N_2]/dt = 1 \times 0.30 =$ **$0.30\ mol\ L^{-1}\ s^{-1}$**

Check: H_2 (coefficient 3) disappears 3 times faster than N_2 (coefficient 1). ✓

2 Rate Law, Order & Rate Constant

The equation that links rate to concentration

For a reaction aA + bB → products, the **rate law** (rate equation) is:

$$\text{Rate} = k[A]^m[B]^n$$

- **k** = **rate constant**: a number specific to the reaction at a given temperature.
- **m** and **n** = **orders** with respect to A and B. They can be 0, 1, 2, fractions, even negative.
- **Overall order** = m + n.
- **m and n are found by experiment.** They are **not** the coefficients a and b (except for a true one-step elementary reaction).

EXAM TRAP **The #1 trap in this chapter**

For $2NO + O_2 \rightarrow 2NO_2$ the rate law is **rate = k[NO]^2[O$_2$]**. Here it matches the coefficients, but that is a coincidence of the mechanism. For $NO_2 + CO \rightarrow NO + CO_2$ the rate law is **rate = k[NO$_2$]^2**: CO does not appear at all. **Never read the order off the balanced equation.**

What order means in practice

Order in A	Rate law	If [A] is doubled, rate…	If [A] is tripled, rate…	Meaning
0 (zero)	rate = k	stays the same (×1)	stays the same (×1)	Rate does not depend on [A]
1 (first)	rate = k[A]	doubles (×2)	triples (×3)	Rate ∝ [A]
2 (second)	rate = k[A]^2	quadruples (×4)	×9	Rate ∝ [A]^2
3 (third)	rate = k[A]^3	×8	×27	Rate ∝ [A]^3

MNEMONIC **Double it and look: 1 – 2 – 4 – 8**

Double [A]: rate ×1 → zero order | ×2 → first | ×4 → second | ×8 → third

General rule: rate factor = (concentration factor)^{order}. So 2^0=1, 2^1=2, 2^2=4, 2^3=8.

The rate constant, k

- Large k = fast reaction; small k = slow reaction (at the same temperature and concentrations).
- k **depends on temperature** (increases with T) and on the **presence of a catalyst**.
- k does **not** depend on concentration. Changing [A] changes the *rate*, not k.
- The **units of k change with the overall order** (below).

★ HIGH-YIELD **Units of k (examine this every year)**

Start from k = rate / (concentration)^n and substitute units (rate in mol L^{-1} s^{-1}, concentration in mol L^{-1}).

Overall order n	Rate law example	Units of k	Written as
0	rate = k	mol L^{-1} s^{-1}	mol dm^{-3} s^{-1}
1	rate = k[A]	s^{-1}	per second
2	rate = k[A]^2 or k[A][B]	L mol^{-1} s^{-1}	dm^3 mol^{-1} s^{-1}
3	rate = k[A]^2[B]	L^2 mol^{-2} s^{-1}	dm^6 mol^{-2} s^{-1}

MNEMONIC **Units of k, the easy way**

Order 0: k has the units of RATE. Each step up in order multiplies by "L per mol".

mol L^{-1} s^{-1} → × L mol^{-1} → s^{-1} → × L mol^{-1} → L mol^{-1} s^{-1} → × L mol^{-1} → L^2 mol^{-2} s^{-1}.

General formula: units of k = (mol L^{-1})$^{(1 - n)}$ s^{-1}

BROKEN DOWN **Molecularity vs order**

Order comes from experiment and describes the **overall rate law**. It can be 0, a fraction or negative.

Molecularity is the number of particles colliding in a **single elementary step**: **1** (unimolecular), **2** (bimolecular), **3** (termolecular, rare). It is always a whole number, never 0 or a fraction, and it is only defined for elementary steps (not for an overall multistep reaction).

3 Finding the Order from Data

Method of initial rates, graphs and half-lives

To find the rate law you need experiments. The standard exam method compares the **initial rates** of runs where the starting concentrations differ.

★ **HIGH-YIELD** **Method of initial rates: the recipe**

- **1.** Pick **two experiments where only ONE concentration changes** (everything else identical).
- **2.** Divide: $rate_2$ / $rate_1$ = $([A]_2 / [A]_1)^m$. Solve for m (use the 1-2-4-8 trick or logs).
- **3.** Repeat for each reactant. Write the rate law.
- **4.** Put **any one** experiment's numbers into the rate law and solve for **k**; add its **units**.
- **5.** (If asked) Use the full rate law to predict the rate for new concentrations.

WORKED EXAMPLE **3: Initial rates ($2NO + O_2 \rightarrow 2NO_2$)**

Find the rate law, the value of k with units, and the overall order.

Exp	[NO] (M)	[O_2] (M)	Initial rate (M s^{-1})
1	0.010	0.010	2.5×10^{-5}
2	0.020	0.010	1.0×10^{-4}
3	0.010	0.020	5.0×10^{-5}

Order in NO (Exp 1 → 2: [NO] doubles, [O_2] constant): rate goes 2.5×10^{-5} → 1.0×10^{-4} = **×4**. So $2^m = 4$, **m = 2**.

Order in O_2 (Exp 1 → 3: [O_2] doubles, [NO] constant): rate ×2. So $2^n = 2$, **n = 1**.

Rate law: rate = $k[NO]^2[O_2]$ → **overall order = 3**.

k (use Exp 1): k = rate / ($[NO]^2[O_2]$) = 2.5×10^{-5} / $[(0.010)^2(0.010)]$ = 2.5×10^{-5} / 1.0×10^{-6} = **25 L^2 mol^{-2} s^{-1}**

Predict: at [NO] = 0.030 M and [O_2] = 0.020 M: rate = 25 $(0.030)^2(0.020)$ = 25 × 9.0×10^{-4} × 0.020 = **4.5×10^{-4} M s^{-1}**

WORKED EXAMPLE 4: A reactant with zero order (A + B → products)

Find the rate law and k.

Exp	[A] (M)	[B] (M)	Initial rate (M s^{-1})
1	0.10	0.10	3.0×10^{-3}
2	0.30	0.10	9.0×10^{-3}
3	0.10	0.40	3.0×10^{-3}

A (Exp 1 → 2: [A] ×3, rate ×3): $3^m = 3$, so **m = 1**.

B (Exp 1 → 3: [B] ×4, rate ×1 i.e. unchanged): $4^n = 1$, so **n = 0**. B has no effect on the rate.

Rate law: rate = k[A]. Overall order 1. k = 3.0×10^{-3} / 0.10 = **3.0×10^{-2} s^{-1}**

EXAM TRAP Initial-rates traps

- Using two experiments where **both** concentrations change. You cannot separate the effects.
- Forgetting the **units** on k, or using the wrong ones for the order.
- If a concentration change gives ×1 for the rate, the order is **0** (not "no answer").

Other ways to find the order

- **Graphical method:** plot [A], ln[A] and 1/[A] against time; the one that gives a **straight line** tells you the order (see Part 4).
- **Half-life method:** constant half-life means first order; half-life that doubles each time means second order; half-life that shrinks as [A] falls means zero order.
- **Differential method:** plot log(rate) against log[A]; the slope equals the order.

WORKED EXAMPLE 5: Which graph is straight? (graphical method)

Concentration of A with time (t in min):

t (min)	0	10	20	30
[A] (M)	1.00	0.607	0.368	0.223
ln[A]	0.00	−0.50	−1.00	−1.50
1/[A] (M^{-1})	1.00	1.65	2.72	4.48

[A] drops by 0.393, then 0.239, then 0.145: **not constant**, so not zero order. 1/[A] rises by 0.65, 1.07, 1.76: **not constant**, so not second order.

ln[A] falls by exactly **0.50 every 10 min**: a straight line. The reaction is **first order** with slope = −k = −0.50/10, so **k = 0.050 min^{-1}**.

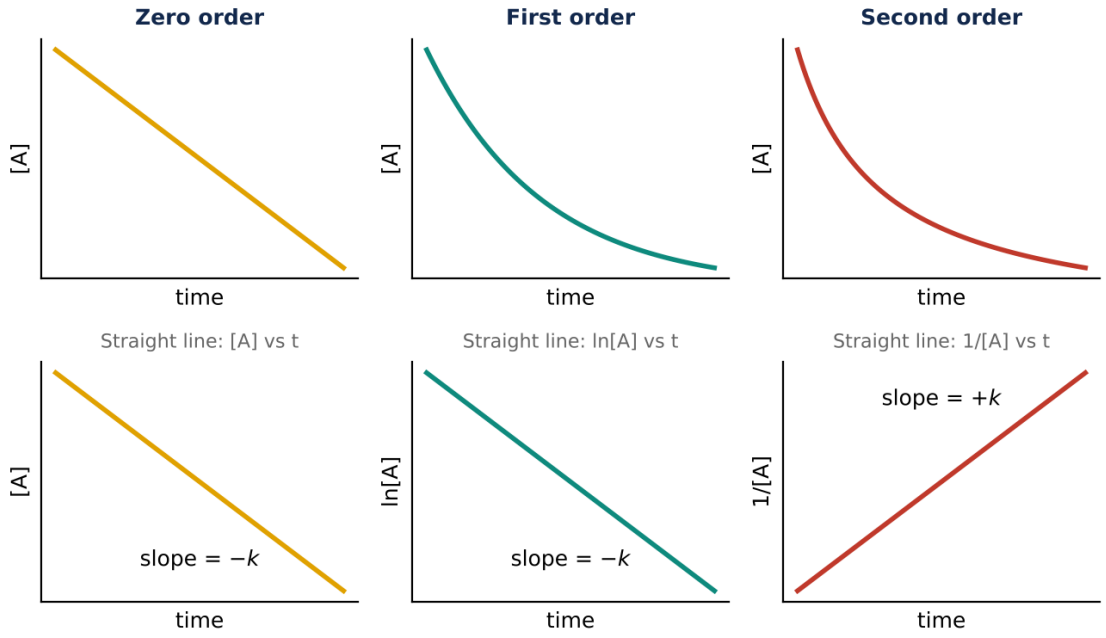
4 Integrated Rate Laws & Half-Life

How concentration changes with time (the calculation heart of the chapter)

The rate law tells you rate vs *concentration*. Integrating it gives **concentration vs time**, which lets you find how much is left after a given time, or how long a given change takes.

★ HIGH-YIELD **Master table: zero, first and second order**

	Zero order	First order	Second order
Rate law	rate = k	rate = k[A]	rate = $k[A]^2$
Integrated law	$[A] = [A]_0 - kt$	$\ln[A] = \ln[A]_0 - kt$ or $[A] = [A]_0 e^{-kt}$	$1/[A] = 1/[A]_0 + kt$
Straight-line plot	[A] vs t	ln[A] vs t	1/[A] vs t
Slope	−k	−k	+k
Half-life $t_{½}$	$[A]_0 / 2k$	0.693 / k	$1 / (k[A]_0)$
$t_{½}$ vs $[A]_0$	directly proportional	independent	inversely proportional
Units of k	mol L^{-1} s^{-1}	s^{-1}	L mol^{-1} s^{-1}



Top row: [A] vs time. Bottom row: the plot that straightens each order.

MNEMONIC Zero Is Absolute, One Is Natural, Two Is Inverse

Zero → [A] (Absolute) | One → ln[A] (Natural log) | Two → 1/[A] (Inverse)

Plot that quantity against time and you get the straight line. **Slopes:** “Zero and One fall, Two climbs”: −k, −k, +k (1/[A] grows as A is used up).

MNEMONIC Half-life dependence on starting concentration

Zero’s Direct, One’s Done (constant), Two’s Inverse

Zero order: more starting material → longer half-life. First order: half-life never changes. Second order: more starting material → shorter half-life, and each successive half-life doubles.

BROKEN DOWN Where the first-order equation comes from (5 steps)

- **1.** Rate = $-d[A]/dt = k[A]$.
- **2.** Separate the variables: $d[A]/[A] = -k\,dt$.
- **3.** Integrate from $[A]_0$ (at t = 0) to [A] (at time t): $\ln[A] - \ln[A]_0 = -kt$.
- **4.** Rearrange: $\ln([A]/[A]_0) = -kt$, or $[A] = [A]_0 e^{-kt}$. Base-10 version: $\log([A]_0/[A]) = kt / 2.303$.
- **5.** Half-life: put $[A] = [A]_0/2$: $\ln(½) = -k\,t_{½} \rightarrow$ **$t_{½} = \ln 2 / k = 0.693 / k$**.

Half-life and “fraction remaining”

Half-lives elapsed (n)	0	1	2	3	4	5
Fraction remaining $(½)^n$	1	1/2	1/4	1/8	1/16	1/32
% remaining	100	50	25	12.5	6.25	3.125
% reacted	0	50	75	87.5	93.75	96.875

MNEMONIC Halve it, n times

100 → 50 → 25 → 12.5 → 6.25 → 3.1 (that is 0 to 5 half-lives)

Shortcut for exams: work out how many half-lives fit into the time, then halve that many times. Only works directly for **first-order** processes (constant half-life), which includes **radioactive decay** and drug elimination.

WORKED EXAMPLE 6: First-order calculations

N_2O_5 decomposes by a first-order process with k = **5.0×10^{-4} s^{-1}**. Starting concentration = 0.40 M.

(a) [N_2O_5] after 1000 s: $\ln[A] = \ln(0.40) - (5.0\times10^{-4})(1000) = -0.916 - 0.500 = -1.416 \rightarrow [A] = e^{-1.416} =$ **0.243 M**. (Or: $0.40 \times e^{-0.5} = 0.40 \times 0.6065 = 0.243$.)

(b) Half-life: $t_{½} = 0.693 / 5.0\times10^{-4} =$ **1.39×10^3 s** (≈ 23.1 min).

(c) Time for 90% to react ([A] = 0.10 $[A]_0$): $t = (1/k)\ln([A]_0/[A]) = \ln(10) / 5.0\times10^{-4} = 2.303 / 5.0\times10^{-4}$ = **4.6×10^3 s**.

WORKED EXAMPLE 7: Half-life counting

A first-order reaction has $t_{½}$ = 20 min. Starting with 64 g:

(a) Mass left after 80 min: 80/20 = 4 half-lives → 64 → 32 → 16 → 8 → **4 g**.

(b) Time for 87.5% to react: 87.5% reacted = 12.5% left = $(½)^3$ → 3 half-lives = **60 min**.

(c) k = 0.693 / 20 min = **0.0347 min^{-1}**.

WORKED EXAMPLE 8: Second-order

A second-order reaction has k = **0.020 L mol^{-1} s^{-1}** and $[A]_0$ = **0.50 M**.

(a) Half-life: $t_{½} = 1/(k[A]_0) = 1/(0.020 \times 0.50) =$ **100 s**.

(b) [A] after 300 s: $1/[A] = 1/[A]_0 + kt = 1/0.50 + (0.020)(300) = 2 + 6 = 8\ M^{-1} \rightarrow [A] =$ **0.125 M**.

Sanity check: first half-life 100 s (0.50 → 0.25), the next is **twice as long**, 200 s (0.25 → 0.125). Total 300 s. ✓

WORKED EXAMPLE 9: Zero-order

A zero-order reaction has k = **5.0×10^{-3} mol L^{-1} s^{-1}**, $[A]_0$ = **0.80 M**.

(a) [A] after 50 s = $0.80 - (5.0\times10^{-3})(50) = 0.80 - 0.25 =$ **0.55 M**.

(b) Half-life = $[A]_0 / 2k = 0.80 / (2 \times 5.0\times10^{-3}) =$ **80 s**.

(c) Time to use up all A = $[A]_0 / k = 0.80 / 5.0\times10^{-3} =$ **160 s**. (Only zero order reaches exactly zero in finite time.)

BROKEN DOWN **Pseudo-first-order reactions**

If one reactant is in **huge excess** (like water as solvent), its concentration barely changes, so it folds into the constant: rate = k[A][B] → rate = k'[A], with k' = k[B]. It **behaves** first order.

Classic examples: hydrolysis of an ester in dilute aqueous solution, and inversion of sucrose in water.

5 Collision Theory & Factors Affecting Rate

Why reactions happen at the molecular level

Collision theory says a reaction occurs only when reactant particles **collide**, but most collisions do not result in a reaction. For a collision to be **effective** it needs three things:

★ **HIGH-YIELD** **The three conditions for an effective collision**

CEO: Collision, Energy, Orientation

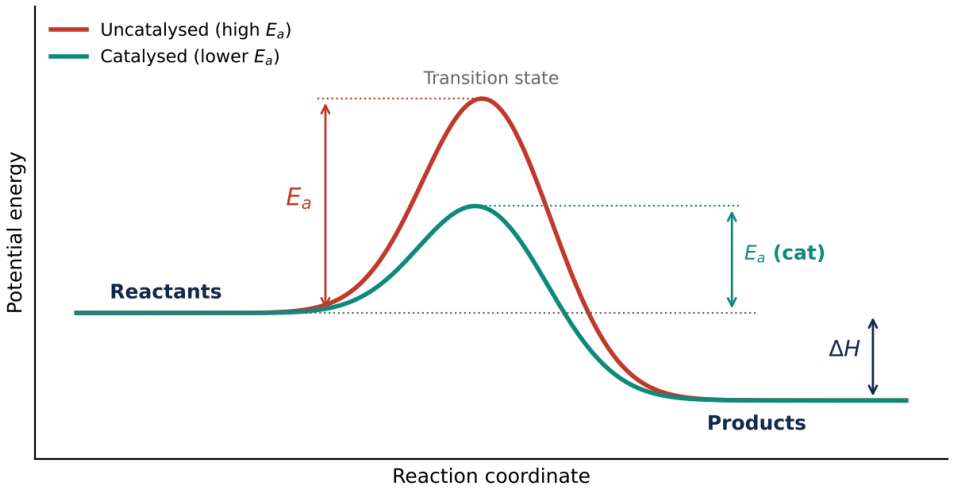
C – Collision: particles must meet.

E – Energy: the colliding particles must have kinetic energy ≥ **activation energy (E_a)**.

O – Orientation: they must hit in the right geometry so the reacting parts meet (the **steric factor**).

Rate ∝ (collision frequency) × (fraction with $E \geq E_a$) × (fraction with correct orientation).

Activation energy (E_a) is the **minimum energy** colliding particles must have for a reaction to occur. It is the energy barrier between reactants and the **transition state** (activated complex), the unstable highest-energy arrangement on the way to products.



Energy profile (exothermic reaction): a catalyst lowers E_a but ΔH is unchanged.

★ **HIGH-YIELD** **Reading an energy profile**

- **E_a (forward)** = peak − reactants. **E_a (reverse)** = peak − products.
- **ΔH** = energy of products − energy of reactants = **E_a(forward) − E_a(reverse)**.
- **Exothermic:** products lower than reactants (ΔH negative). **Endothermic:** products higher (ΔH positive).
- The peak is the **transition state**, not an intermediate (an intermediate sits in a dip between two peaks).

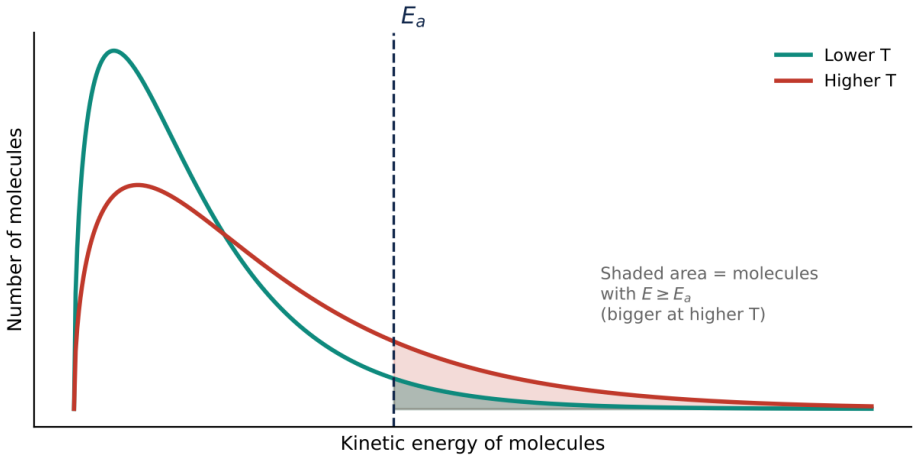
Factors affecting reaction rate

MNEMONIC **SCTP-NCL**

Smart Chemists Take Pills, Never Cry Loudly

Surface area • **C**oncentration • **T**emperature • **P**ressure (gases) • **N**ature of reactants • **C**atalyst • **L**ight

Factor	Effect on rate	Explanation	Changes k?
Concentration	Higher → faster	More particles per volume → more frequent collisions	No
Pressure (gases)	Higher → faster	Same as raising concentration (particles squeezed closer)	No
Surface area	Larger → faster	More exposed particles for solids; powder reacts faster than a lump (heterogeneous reactions)	No
Temperature	Higher → much faster	More molecules have $E \geq E_a$ AND collide more often (the first effect is the big one)	**Yes**
Catalyst	Faster	Provides an alternative pathway with lower E_a	**Yes**
Nature of reactants	Varies	Ionic reactions in solution are fast (no bonds to break); covalent reactions are slower (bonds must be broken)	Yes (different reaction)
Light	Faster for photochemical reactions	Photons supply the activation energy (e.g. $H_2 + Cl_2$, photosynthesis, AgBr in photography)	Yes



Maxwell–Boltzmann distribution: a small rise in T puts far more molecules above E_a.

BROKEN DOWN Why a small temperature rise has a big effect

The average energy barely changes, but the **tail** of the distribution (molecules with $E \geq E_a$) grows very quickly. That is why rate roughly **doubles for every 10 °C** rise near room temperature (a rule of thumb, not an exact law).

Concentration raises the rate by increasing collision frequency, but it does not change the fraction of molecules with enough energy, so k stays the same.

EXAM TRAP Which factors change k?

Only **temperature** and **catalyst** (and the nature of the reaction itself). Concentration, pressure and surface area change the **rate** but **not k**.

6 Activation Energy & the Arrhenius Equation

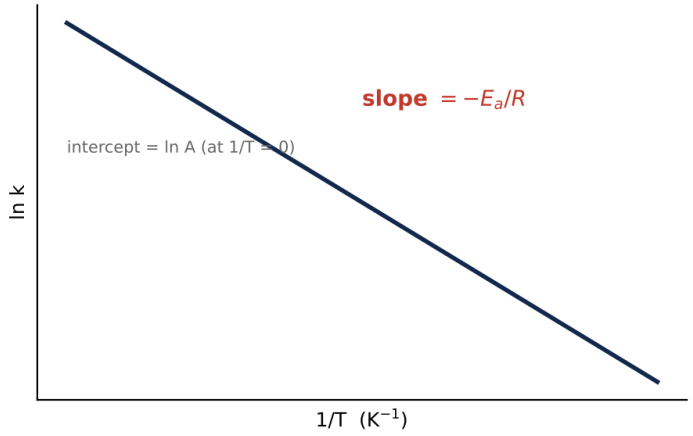
How k depends on temperature

$$k = A\, e^{-E_a/RT}$$

- **k** = rate constant | **A** = frequency (pre-exponential) factor: collision frequency × orientation factor
- **E_a** = activation energy (J mol^{-1}) | **R** = 8.314 J mol^{-1} K^{-1} | **T** = temperature in **kelvin**
- **$e^{-E_a/RT}$** = the fraction of collisions with energy ≥ E_a. Higher E_a → smaller fraction → smaller k. Higher T → larger fraction → larger k.

Take natural logs to get a **straight-line form**:

$$\ln k = \ln A - (E_a/R)(1/T)$$



Arrhenius plot: ln k against 1/T gives a straight line.

MNEMONIC **Arrhenius slope**

Slope = -Ea / R (“Slope Equals Minus Energy over R”)

Steeper line → bigger E_a → rate more sensitive to temperature. **E_a = −slope × R.**

Two-temperature form (very common in exams)

$$\ln(k_2 / k_1) = (E_a / R)(1/T_1 - 1/T_2)$$

★ **HIGH-YIELD** **Arrhenius checklist**

- Convert **°C to K** (add 273 or 273.15) before anything else.
- R = 8.314 **J** mol^{-1} K^{-1}, so E_a comes out in **J mol^{-1}**. Convert to kJ if asked (÷ 1000), or use E_a in J when substituting.
- Keep the order straight: T_2 is the higher temperature, k_2 the larger k. Then $(1/T_1 - 1/T_2)$ is positive.

WORKED EXAMPLE **10: Find E_a from two temperatures**

k = 2.0×10^{-3} s^{-1} at 300 K and 4.0×10^{-3} s^{-1} at 310 K. Find E_a.

$\ln(k_2/k_1) = \ln 2 = 0.693$. $1/T_1 - 1/T_2 = 1/300 - 1/310 = 1.075 \times 10^{-4}$ K^{-1}

$E_a = 0.693 \times 8.314 / 1.075\times10^{-4} = 5.36 \times 10^4$ J mol^{-1} = **53.6 kJ mol^{-1}**.

Check: a rate that doubles for a 10 K rise at room temperature corresponds to E_a of roughly 50 kJ mol^{-1}. ✓

WORKED EXAMPLE 11: Find the new rate constant

E_a = 50.0 kJ mol^{-1} and k = 1.0×10^{-3} s^{-1} at 298 K. Find k at 318 K.

$\ln(k_2/k_1) = (50\,000 / 8.314)(1/298 - 1/318) = 6014 \times 2.11\times10^{-4} = 1.269$

$k_2/k_1 = e^{1.269} = 3.56 \rightarrow$ **k_2 = 3.6 × 10^{-3} s^{-1}** (about 3.6 times faster for a 20 K rise).

WORKED EXAMPLE 12: E_a from an Arrhenius graph

A plot of ln k against 1/T has slope **−6.0 × 10^3 K**. E_a = −slope × R = 6.0×10^3 × 8.314 = 4.99 × 10^4 J mol^{-1} ≈ **49.9 kJ mol^{-1}**.

7 Catalysis & Enzymes

Speeding reactions up without being used up

A **catalyst** is a substance that **increases the rate** of a reaction by providing an **alternative pathway with a lower activation energy**, and is **not consumed** (it is recovered unchanged at the end).

★ **HIGH-YIELD What a catalyst does and does not do**

- **DOES:** lower E_a (alternative route); speed up the **forward and reverse** reactions equally; help equilibrium be reached **faster**.
- **DOES NOT:** change ΔH, change the position of equilibrium or the value of K, change the yield at equilibrium, or make a non-spontaneous reaction spontaneous.
- Is needed only in **small amounts**; is often **specific** to a particular reaction; can be **poisoned** by impurities.

MNEMONIC Catalyst = tour guide

Cheaper route, same destination.

Same start (reactants), same finish (products), same energy difference (ΔH), but a lower mountain pass (E_a) in between.

Types of catalysis

Type	Phase	Examples
Homogeneous	Same phase as the reactants	H^+ catalysing ester hydrolysis; NO gas catalysing O_3 breakdown; Cl radicals and the ozone layer
Heterogeneous	Different phase (usually a solid catalyst, gas/liquid reactants)	Fe in the **Haber** process (N_2 + H_2); V_2O_5 in the **Contact** process; Ni in hydrogenation of oils; Pt/Rh in catalytic converters; MnO_2 in $KClO_3$ decomposition
Enzyme (bio)catalysis	Proteins in living systems	Amylase (starch), catalase ($H_2O_2 \rightarrow H_2O + O_2$), urease, lipases
Autocatalysis	A product acts as the catalyst	Mn^{2+} formed in the $KMnO_4$ / oxalate reaction: slow at first, then speeds up

BROKEN DOWN How a solid (heterogeneous) catalyst works

- **1.** Reactant molecules are **adsorbed** onto active sites on the surface.
- **2.** Bonds in the reactants are weakened and the molecules are held in a favourable orientation.
- **3.** Reaction occurs on the surface. **4.** Products are **desorbed**, freeing the site for more reactant.

Enzymes (important for health-science students)

- Enzymes are **biological catalysts** (proteins). The reactant is the **substrate**; it binds at the **active site** to form an enzyme–substrate complex (lock-and-key / induced-fit model).

- Very **specific** and very efficient. Work best at an **optimum temperature** (about 37 °C in humans) and **optimum pH**. Too hot or wrong pH → the protein **denatures** and activity is lost.
- **Inhibitors** slow enzyme action. Rate increases with [substrate] and then **levels off (saturation)** when all active sites are occupied; at that point the rate is effectively zero order in substrate.

EXAM TRAP **Inhibitor vs poison vs catalyst**

A **negative catalyst (inhibitor)** raises E_a or blocks a pathway to slow a reaction. A **catalyst poison** permanently blocks the active sites of a catalyst (e.g. lead in a catalytic converter). A catalyst **never changes ΔH or K**.

8 Reaction Mechanisms

The step-by-step story behind the rate law

Most reactions happen in a **sequence of simple steps** called a **mechanism**. Each step is an **elementary reaction** (a single molecular event), and for an elementary step **the rate law *can* be read straight from the equation**.

Elementary step	Molecularity	Rate law
A → products	Unimolecular	rate = k[A]
A + A → products or A + B → products	Bimolecular	rate = $k[A]^2$ or k[A][B]
A + B + C → products (rare)	Termolecular	rate = k[A][B][C]

★ **HIGH-YIELD** **Rules for mechanisms**

- The elementary steps must **add up to the overall balanced equation** (species that appear on both sides cancel).
- The **slowest step is the rate-determining step (RDS)**. The overall rate law is the rate law of the slow step.
- A proposed mechanism must **agree with the experimental rate law**. A mechanism can be disproved by experiment but never proved.
- **Intermediate** = formed in one step and used up in a later step (never in the overall equation). **Catalyst** = used in one step and regenerated later.

MNEMONIC **Intermediate vs catalyst**

Intermediate: made first, used later. Catalyst: used first, remade later.

Think of a bottleneck on a road: the whole journey is only as fast as the slowest section (RDS).

WORKED EXAMPLE **13: Mechanism and rate law**

Overall reaction: **$NO_2 + CO \rightarrow NO + CO_2$**. Experimental rate law: **rate = $k[NO_2]^2$**.

Proposed mechanism:

Step 1 (slow): $NO_2 + NO_2 \rightarrow NO_3 + NO$

Step 2 (fast): $NO_3 + CO \rightarrow NO_2 + CO_2$

Sum check: $2NO_2 + NO_3 + CO \rightarrow NO_3 + NO + NO_2 + CO_2$; cancel NO_3 and one NO_2 → $NO_2 + CO \rightarrow NO + CO_2$. ✓

Rate law from the slow step: rate = $k[NO_2][NO_2]$ = **$k[NO_2]^2$**. Matches the experiment. ✓

NO_3 is an intermediate (made in step 1, used in step 2). CO appears only in the fast step, so it does not appear in the rate law.

WORKED EXAMPLE 14: Fast equilibrium first (still examinable)

Overall: $2NO + O_2 \rightarrow 2NO_2$. Experimental rate = $k[NO]^2[O_2]$.

Step 1 (fast, reversible): $NO + O_2 \rightleftharpoons NO_3$ Step 2 (slow): $NO_3 + NO \rightarrow 2NO_2$

Slow step gives: rate = $k_2[NO_3][NO]$, but NO_3 is an intermediate and cannot appear in the final law.

From the fast equilibrium: $[NO_3] = K[NO][O_2]$. Substitute: rate = $k_2K[NO]^2[O_2]$ = **$k'[NO]^2[O_2]$**. Matches. ✓

9 Exam Traps & Common Mistakes

Check this list before you write your final answer

Mistake	Correct approach
Reading the order from the balanced equation	Orders come from **experiment** (or the slow step of a mechanism).
Confusing order with molecularity	Order: experimental, any value. Molecularity: elementary step only, whole number 1, 2 or 3.
Wrong units for k	Use the "L per mol each step up" rule: order 0 → mol L^{-1} s^{-1}; 1 → s^{-1}; 2 → L mol^{-1} s^{-1}
Using °C in Arrhenius	Always **kelvin**.
Using ln when the question gives log (or vice versa)	ln x = 2.303 log x. Be consistent and use the right calculator key.
Mixing up the slope sign	Zero and first order slope = −k; second order slope = +k; Arrhenius slope = $-E_a/R$.
Saying a catalyst "changes ΔH" or "shifts equilibrium"	It lowers E_a only. ΔH, K and equilibrium yield are unchanged.
Thinking concentration changes k	It changes the **rate**. Only temperature (and catalyst) change **k**.
Using the first-order half-life formula for every order	0.693/k is for **first order only**. Zero: $[A]_0/2k$. Second: $1/(k[A]_0)$.
Forgetting to divide by the coefficient	Reaction rate = (1/coefficient) × d[X]/dt for any species.
Calling the transition state an intermediate	Transition state = peak (cannot be isolated). Intermediate = dip between peaks (can sometimes be detected).
Doing initial-rates with two changing concentrations	Change **one** concentration at a time.

10 Key Definitions (Exam-Ready)

Short, mark-scoring wording

Term	Definition
Chemical kinetics	The study of the rates of chemical reactions, the factors affecting them and the mechanisms by which they occur.
Rate of reaction	The change in concentration of a reactant or product per unit time.
Rate law	An equation relating the rate of a reaction to the concentrations of reactants raised to powers (orders), determined experimentally.
Order of reaction	The power to which the concentration of a reactant is raised in the rate law. The overall order is the sum of the powers.
Rate constant (k)	The proportionality constant in the rate law; it is constant at a given temperature and depends on temperature and catalyst.
Half-life ($t_{½}$)	The time taken for the concentration of a reactant to fall to half of its initial value.
Activation energy (E_a)	The minimum energy colliding particles must possess for a reaction to occur.
Catalyst	A substance that increases the rate of a reaction by providing an alternative pathway of lower activation energy, and is not consumed in the process.
Elementary reaction	A single-step reaction that occurs exactly as written, in one molecular event.
Molecularity	The number of reacting particles that collide in an elementary step.
Rate-determining step	The slowest step in a reaction mechanism; it controls the overall rate.
Intermediate	A species formed in one step of a mechanism and consumed in a later step; it does not appear in the overall equation.
Transition state	The highest-energy, unstable arrangement of atoms at the top of the energy barrier between reactants and products.
Effective collision	A collision with enough energy ($\geq E_a$) and the correct orientation to cause a reaction.

11 One-Page Formula Sheet

Screenshot this and keep it on your phone

Quantity	Formula
Reaction rate	$\text{Rate} = -(1/a)d[A]/dt = +(1/c)d[C]/dt$
Rate law	$\text{Rate} = k[A]^m[B]^n$ (overall order = m + n)
Order from two experiments	$\text{rate}_2/\text{rate}_1 = ([A]_2/[A]_1)^m$
Units of k	$(\text{mol L}^{-1})^{(1-n)}\ \text{s}^{-1}$
Zero order	$[A] = [A]_0 - kt$ $t_{½} = [A]_0/2k$
First order	$\ln[A] = \ln[A]_0 - kt$ $[A] = [A]_0e^{-kt}$ $t_{½} = 0.693/k$ $\log([A]_0/[A]) = kt/2.303$
Second order	$1/[A] = 1/[A]_0 + kt$ $t_{½} = 1/(k[A]_0)$
Fraction left after n half-lives	$(½)^n$ with $n = t / t_{½}$
Arrhenius	$k = Ae^{-E_a/RT}$ $\ln k = \ln A - E_a/RT$
Arrhenius (two temperatures)	$\ln(k_2/k_1) = (E_a/R)(1/T_1 - 1/T_2)$
Enthalpy from energy profile	$\Delta H = E_a(\text{forward}) - E_a(\text{reverse})$
Constants	$R = 8.314\ \text{J mol}^{-1}\ \text{K}^{-1}$ $\ln 2 = 0.693$ $\ln 10 = 2.303$ $T(\text{K}) = °C + 273$

MNEMONIC **Memory pack (all in one place)**

- **Sign & coefficient:** Reactants Reduce (−), Products Pile up (+).
- **Order from data:** 1 – 2 – 4 – 8 for orders 0 – 1 – 2 – 3.
- **Units of k:** order 0 = units of rate; each step up multiplies by L mol^{-1}.
- **Straight lines:** Zero Is Absolute, One Is Natural, Two Is Inverse ([A], ln[A], 1/[A]).
- **Half-life:** Zero's Direct, One's Done, Two's Inverse.
- **Factors:** Smart Chemists Take Pills, Never Cry Loudly.
- **Collisions:** CEO = Collision, Energy, Orientation.
- **Arrhenius:** Slope Equals Minus Energy over R.
- **Catalyst:** Cheaper route, same destination.
- **Intermediate vs catalyst:** made first, used later vs used first, remade later.

12 Practice Questions

Try each one first, then check the answers on the next page

Q1. State the units of the rate constant for (a) a first-order and (b) a second-order reaction.

Q2. For 2A + 3B → C, the rate of disappearance of A is 0.040 mol L^{-1} s^{-1}. Find the rate of disappearance of B and the rate of formation of C.

Q3. The rate of a reaction doubles when [A] is doubled and quadruples when [B] is doubled. Write the rate law and the overall order. By what factor does the rate change if both [A] and [B] are tripled?

Q4. A first-order reaction has k = 0.0231 min^{-1}. Calculate the half-life and the time taken for 75% of the reactant to be consumed.

Q5. A reaction has a half-life of 50 s when $[A]_0$ = 1.0 M and 100 s when $[A]_0$ = 2.0 M. What is the order? Why?

Q6. Which plot is a straight line for a second-order reaction, and what is its slope?

Q7. A graph of ln k against 1/T gives a straight line of slope -7.5×10^3 K. Calculate E_a (R = 8.314 J mol^{-1} K^{-1}).

Q8. State two effects a catalyst has on a reaction and two things it does NOT change.

Q9. In the mechanism Step 1 (slow): A + B → C; Step 2 (fast): C + B → D, write the overall equation, the rate law, and identify the intermediate.

Q10. The first-order decay of a drug in the body has $t_{½}$ = 6 h. What fraction of a dose remains after 24 h?

Answers

A1. (a) $\mathbf{s^{-1}}$ (b) $\mathbf{L\ mol^{-1}\ s^{-1}}$.

A2. Reaction rate = 0.040/2 = 0.020. B: 3 × 0.020 = $\mathbf{0.060\ mol\ L^{-1}\ s^{-1}}$ (disappearance). C: $\mathbf{0.020\ mol\ L^{-1}\ s^{-1}}$ (formation).

A3. Rate = $\mathbf{k[A][B]^2}$, overall order **3**. Tripling both: $3^1 \times 3^2$ = **27 times**.

A4. $t_{½}$ = 0.693/0.0231 = **30 min**. 75% consumed = 25% left = 2 half-lives = **60 min**.

A5. Zero order. Half-life is directly proportional to $[A]_0$ (doubling $[A]_0$ doubles $t_{½}$).

A6. 1/[A] against time; slope = **+k**.

A7. E_a = −slope × R = $7.5\times10^3 \times 8.314 = 6.24 \times 10^4$ J mol^{-1} = $\mathbf{62.4\ kJ\ mol^{-1}}$.

A8. Does: lowers E_a by an alternative pathway; increases the rate of forward and reverse reactions (reaches equilibrium faster). **Does not change:** ΔH, equilibrium constant K / position of equilibrium (and it is not consumed).

A9. Overall: **A + 2B → D**. Rate law from the slow step: **rate = k[A][B]**. Intermediate: **C**.

A10. 24/6 = 4 half-lives: $(½)^4$ = **1/16** (6.25%) remains.

YOU HAVE JUST COVERED THE WHOLE CHAPTER.

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